

School of Computer Science and Artificial Intelligence

WEEK-9

Assignment #17

Program : B. Tech (CSE)

Specialization :AIML

Course Title : Competitive Programming

Course Code : 23CS302PC305

Semester : VI

Academic Session : 2025-2026

Name of Student : Amma Anjali

Enrollment No. : 2303A52385

Batch No. : 33

QUESTION:

Search Engine Query Processing (Suffix Array)

A search engine indexes a large document to efficiently answer multiple substring queries. To enable fast searches, a Suffix Array is constructed for the document.

Task

1. Construct the suffix array of the given text.
2. Use binary search on the suffix array to process query strings.
3. Report whether each query exists in the text. Example Test Case 1

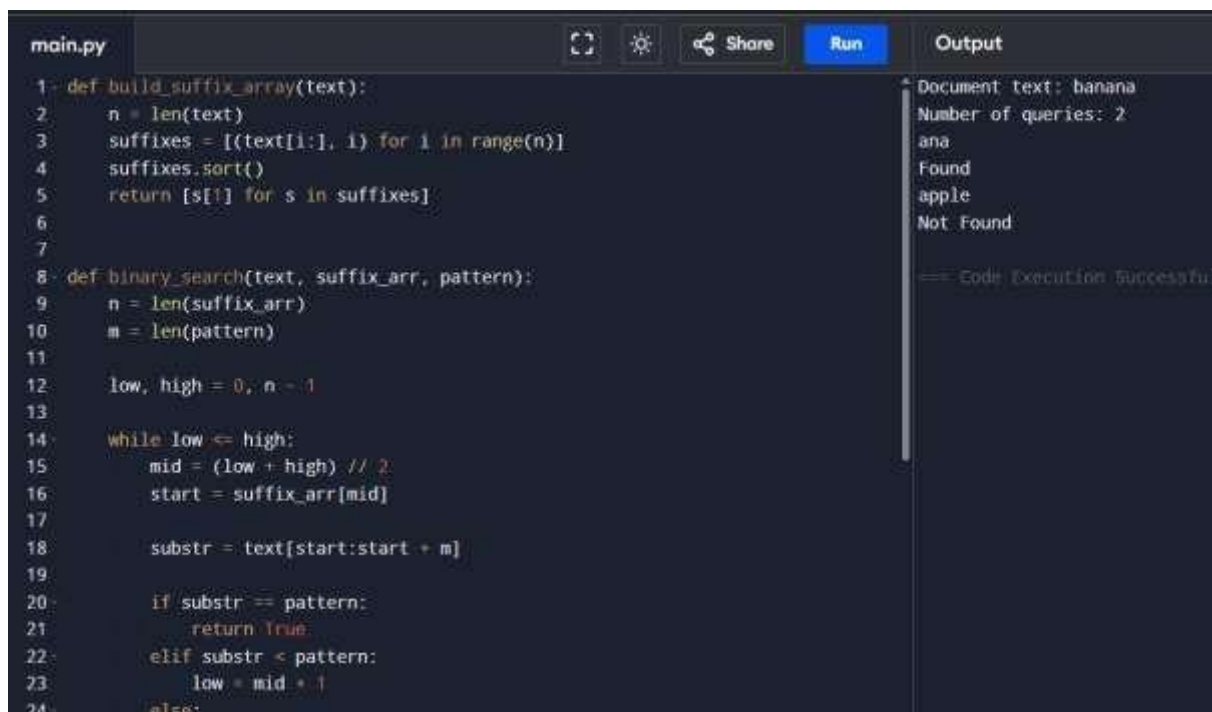
Input

Document text: banana Number of queries: 2

Queries: ana apple Output Found

Not Found

PYTHON CODE:



```
main.py
1 def build_suffix_array(text):
2     n = len(text)
3     suffixes = [(text[i:], i) for i in range(n)]
4     suffixes.sort()
5     return [s[1] for s in suffixes]
6
7
8 def binary_search(text, suffix_arr, pattern):
9     n = len(suffix_arr)
10    m = len(pattern)
11
12    low, high = 0, n - 1
13
14    while low <= high:
15        mid = (low + high) // 2
16        start = suffix_arr[mid]
17
18        substr = text[start:start + m]
19
20        if substr == pattern:
21            return True
22        elif substr < pattern:
23            low = mid + 1
24        else:
```

Document text: banana
Number of queries: 2
ana
Found
apple
Not Found
Code Execution Successful

```
main.py  [Icons]  Share  Run  Output

18     substr = text[start:start + m]
19
20     if substr == pattern:
21         return True
22     elif substr < pattern:
23         low = mid + 1
24     else:
25         high = mid - 1
26
27     return False
28
29
30 # ---- MAIN ----
31 text = input("Document text: ").strip()
32 q = int(input("Number of queries: "))
33
34 suffix_arr = build_suffix_array(text)
35
36 for _ in range(q):
37     pattern = input().strip()
38     if binary_search(text, suffix_arr, pattern):
39         print("Found")
40     else:
41         print("Not Found")
```

Document text: banana
Number of queries: 2
ana
Found
apple
Not Found
== Code Execution Successful ==

JAVA CODE:

```
SuffixArraySearch.java  [Icons]  Share  Run  Output

1- import java.util.*;
2
3- public class SuffixArraySearch {
4
5-     static int[] buildSuffixArray(String text) {
6         int n = text.length();
7         String[] suffixes = new String[n];
8         Integer[] index = new Integer[n];
9
10        for (int i = 0; i < n; i++) {
11            suffixes[i] = text.substring(i);
12            index[i] = i;
13        }
14
15        Arrays.sort(index, (a, b) -> suffixes[a].compareTo(suffixes[b]));
16
17        int[] suffixArr = new int[n];
18        for (int i = 0; i < n; i++) {
19            suffixArr[i] = index[i];
20        }
21
22        return suffixArr;
23    }
24}
```

Document text: banana
Number of queries: 2
ana
Found
apple
Not Found
== Code Execution Successful ==

```
SuffixArraySearch.java  [Icons] Share Run Output
24
25 static boolean binarySearch(String text, int[] suffixArr, String pattern) {
26     int low = 0, high = suffixArr.length - 1;
27     int m = pattern.length();
28
29     while (low <= high) {
30         int mid = (low + high) / 2;
31         int start = suffixArr[mid];
32
33         String substr;
34         if (start + m <= text.length()) {
35             substr = text.substring(start, start + m);
36         } else {
37             substr = text.substring(start);
38         }
39
40         int cmp = substr.compareTo(pattern);
41
42         if (cmp == 0) {
43             return true;
44         } else if (cmp < 0) {
45             low = mid + 1;
46         } else {
47             high = mid - 1;
48     }
}
```

Document text: banana
Number of queries: 2
ana
Found
apple
Not Found
Code Execution Successful

```
SuffixArraySearch.java  [Icons] Share Run Output
53
54 public static void main(String[] args) {
55     Scanner sc = new Scanner(System.in);
56
57     System.out.print("Document text: ");
58     String text = sc.next();
59
60     System.out.print("Number of queries: ");
61     int q = sc.nextInt();
62
63     int[] suffixArr = buildSuffixArray(text);
64
65     for (int i = 0; i < q; i++) {
66         String pattern = sc.next();
67         if (binarySearch(text, suffixArr, pattern)) {
68             System.out.println("Found");
69         } else {
70             System.out.println("Not Found");
71         }
72     }
73
74     sc.close();
75 }
76 }
```

Document text: banana
Number of queries: 2
ana
Found
apple
Not Found
Code Execution Successful

